17 次单位根的根式解

参数与标准素分解

$p = 17$，$h = (p-1)/2 = 8$，模 17 的最小原根 $g = 3$。
h 的标准素分解：$h = 2 \times 2 \times 2$，子群链共 3 层。
$m_0 = 1$，$m_1 = 2$，$m_2 = 4$，$m_3 = 8$。
各层子群 $H_j = \{\pm g^{m_j t}\}$，其中 H_0 是全部 16 个非零剩余：

$$H_1 = \{1,\, 2,\, 4,\, 8,\, 9,\, 13,\, 15,\, 16\}$$

$$H_2 = \{1,\, 4,\, 13,\, 16\}$$

$$H_3 = \{1,\, 16\}$$

各层周期 $\eta_a^{(j)} = \sum_{b \in H_j} \zeta^{ab}$（$a$ 取陪集代表）。

各层周期与分解表

第 0 层：1 个周期

H_0 是全部 16 个非零剩余，故 $\eta_1^{(0)} = -1$。

第 1 层：2 个周期

$$\eta_1^{(1)} = \zeta^1 + \zeta^2 + \zeta^4 + \zeta^8 + \zeta^9 + \zeta^{13} + \zeta^{15} + \zeta^{16}$$

$$\eta_3^{(1)} = \zeta^3 + \zeta^5 + \zeta^6 + \zeta^7 + \zeta^{10} + \zeta^{11} + \zeta^{12} + \zeta^{14}$$

第 2 层：4 个周期

$$\eta_1^{(2)} = \zeta^1 + \zeta^4 + \zeta^{13} + \zeta^{16}$$

$$\eta_2^{(2)} = \zeta^2 + \zeta^8 + \zeta^9 + \zeta^{15}$$

$$\eta_3^{(2)} = \zeta^3 + \zeta^5 + \zeta^{12} + \zeta^{14}$$

$$\eta_6^{(2)} = \zeta^6 + \zeta^7 + \zeta^{10} + \zeta^{11}$$

第 3 层：8 个周期

$$\eta_1^{(3)} = \zeta^{1} + \zeta^{16}$$

$$\eta_2^{(3)} = \zeta^{2} + \zeta^{15}$$

$$\eta_3^{(3)} = \zeta^{3} + \zeta^{14}$$

$$\eta_4^{(3)} = \zeta^{4} + \zeta^{13}$$

$$\eta_5^{(3)} = \zeta^{5} + \zeta^{12}$$

$$\eta_6^{(3)} = \zeta^{6} + \zeta^{11}$$

$$\eta_7^{(3)} = \zeta^{7} + \zeta^{10}$$

$$\eta_8^{(3)} = \zeta^{8} + \zeta^{9}$$

分解表：每个母周期拆成 ℓ_j 个子周期

$$\eta_1^{(0)} = \eta_1^{(1)} + \eta_3^{(1)}$$

（第 1 层，$\ell = 2$。）

$$\eta_1^{(1)} = \eta_1^{(2)} + \eta_2^{(2)}$$

$$\eta_3^{(1)} = \eta_3^{(2)} + \eta_6^{(2)}$$

（第 2 层，$\ell = 2$。）

$$\eta_1^{(2)} = \eta_1^{(3)} + \eta_4^{(3)}$$

$$\eta_2^{(2)} = \eta_2^{(3)} + \eta_8^{(3)}$$

$$\eta_3^{(2)} = \eta_3^{(3)} + \eta_5^{(3)}$$

$$\eta_6^{(2)} = \eta_6^{(3)} + \eta_7^{(3)}$$

（第 3 层，$\ell = 2$。）

辅助单位根

$\ell = 2$ 时 $\omega_2^1 = -1$。

$\ell \geq 3$ 时按主支：

$$\omega_\ell^k = \frac{\Lambda_{\bar{k}}^{(\ell)}}{2} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{\frac{\ell}{2k}}^{(\ell)}}\, i$$

其中 $\varepsilon_k = \operatorname{sgn}\sin(2\pi k/\ell)$，$\bar{k} = \min(k, \ell - k)$。

逐层开方

每一步把母周期 $\eta_a^{(j-1)}$ 分解成 ℓ 个子周期 $\beta_t^{(a)}$。

预解式取离散 Fourier 变换：

$$\theta_{a,k}^{(j)} = \sum_{t=0}^{\ell-1} \omega_\ell^{-kt} \beta_t^{(a)}$$

$$\left(\theta_{a,k}^{(j)}\right)^\ell = u_{a,k}^{(j)} = \tau^k \left(u_{a,1}^{(j)}\right)$$

$$\eta_{b(t)}^{(j)} = \frac{1}{\ell}\left(\eta_a^{(j-1)} + \sum_{k=1}^{\ell-1} \omega_\ell^{kt} \theta_{a,k}^{(j)}\right)$$

（τ 把 ω_ℓ 送到 ω_ℓ^k；$b(t)$ 为第 t 个子周期。）

开方一律取正实主支；预解式的分支由因子 ω_ℓ^s 记录（见报告里的分支表）。

第 1 步（$\ell = 2$，**1 个母周期**）

$$u_{1,1}^{(1)} = 17$$

$$\theta_{1,1}^{(1)} = \sqrt{17}$$

第 2 步（$\ell = 2$，**2 个母周期**）

$$u_{1,1}^{(2)} = \frac{17 - \sqrt{17}}{2}$$

$$\theta_{1,1}^{(2)} = \frac{\left(5 - \sqrt{17}\right)\sqrt{17 + 4\sqrt{17}}}{2}$$

$$u_{3,1}^{(2)} = \frac{17 + \sqrt{17}}{2}$$

$$\theta_{3,1}^{(2)} = \frac{\left(5 + \sqrt{17}\right)\sqrt{17 - 4\sqrt{17}}}{2}$$

第 3 步（$\ell = 2$，**4 个母周期**）

$$u_{1,1}^{(3)} = \frac{17 + 3\sqrt{17} + \left(-5 + \sqrt{17}\right)\sqrt{17 + 4\sqrt{17}} + \left(-10 - 2\sqrt{17}\right)\sqrt{17 - 4\sqrt{17}}}{4}$$

$$\theta_{1,1}^{(3)} = \frac{\sqrt{17 + 3\sqrt{17} + \left(-5 + \sqrt{17}\right)\sqrt{17 + 4\sqrt{17}} + \left(-10 - 2\sqrt{17}\right)\sqrt{17 - 4\sqrt{17}}}}{2}$$

$$u_{2,1}^{(3)} = \frac{17 + 3\sqrt{17} + \left(5 - \sqrt{17}\right)\sqrt{17 + 4\sqrt{17}} + \left(10 + 2\sqrt{17}\right)\sqrt{17 - 4\sqrt{17}}}{4}$$

$$\theta_{2,1}^{(3)} = \frac{\sqrt{17 + 3\sqrt{17} + \left(5 - \sqrt{17}\right)\sqrt{17 + 4\sqrt{17}} + \left(10 + 2\sqrt{17}\right)\sqrt{17 - 4\sqrt{17}}}}{2}$$

$$u_{3,1}^{(3)}=\frac{17-3\sqrt{17}+\left(10-2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-5-\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}{4}$$

$$\theta_{3,1}^{(3)}=\frac{\sqrt{17-3\sqrt{17}+\left(10-2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-5-\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{2}$$

$$u_{6,1}^{(3)}=\frac{17-3\sqrt{17}+\left(-10+2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(5+\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}{4}$$

$$\theta_{6,1}^{(3)}=\frac{\sqrt{17-3\sqrt{17}+\left(-10+2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(5+\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{2}$$

实部数值

$$\Lambda_1=2\cos\frac{2\pi\cdot1}{17}=\frac{-1+\sqrt{17}+\left(5-\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+2\sqrt{17+3\sqrt{17}+\left(-5+\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-10-2\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_2=2\cos\frac{2\pi\cdot2}{17}=\frac{-1+\sqrt{17}+\left(-5+\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+2\sqrt{17+3\sqrt{17}+\left(5-\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(10+2\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_3=2\cos\frac{2\pi\cdot3}{17}=\frac{-1-\sqrt{17}+\left(5+\sqrt{17}\right)\sqrt{17-4\sqrt{17}}+2\sqrt{17-3\sqrt{17}+\left(10-2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-5-\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_4=2\cos\frac{2\pi\cdot4}{17}=\frac{-1+\sqrt{17}+\left(5-\sqrt{17}\right)\sqrt{17+4\sqrt{17}}-2\sqrt{17+3\sqrt{17}+\left(-5+\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-10-2\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_5=2\cos\frac{2\pi\cdot5}{17}=\frac{-1-\sqrt{17}+\left(5+\sqrt{17}\right)\sqrt{17-4\sqrt{17}}-2\sqrt{17-3\sqrt{17}+\left(10-2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-5-\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_6=2\cos\frac{2\pi\cdot6}{17}=\frac{-1-\sqrt{17}+\left(-5-\sqrt{17}\right)\sqrt{17-4\sqrt{17}}+2\sqrt{17-3\sqrt{17}+\left(-10+2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(5+\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_7=2\cos\frac{2\pi\cdot7}{17}=\frac{-1-\sqrt{17}+\left(-5-\sqrt{17}\right)\sqrt{17-4\sqrt{17}}-2\sqrt{17-3\sqrt{17}+\left(-10+2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(5+\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

$$\Lambda_8=2\cos\frac{2\pi\cdot8}{17}=\frac{-1+\sqrt{17}+\left(-5+\sqrt{17}\right)\sqrt{17+4\sqrt{17}}-2\sqrt{17+3\sqrt{17}+\left(5-\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(10+2\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}{8}$$

全部单位根

最终形式：

$$\omega_n^t=\frac{\Lambda_{\bar{t}}}{2}+\frac{\varepsilon_t}{2}\sqrt{2-\Lambda_{\overline{2t}}}\,i,\qquad \Lambda_{\bar{t}}=2\cos\frac{2\pi\bar{t}}{n}$$

其中 $\bar{t}=\min(t,n-t)$，$\varepsilon_t=\operatorname{sgn}\sin(2\pi t/n)$。

以下逐一写开（全部为具体根式）：

$$\omega_{17}^{1}=\frac{-1+\sqrt{17}+\left(5-\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+2\sqrt{17+3\sqrt{17}+\left(-5+\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(-10-2\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}+2i\sqrt{34-2\sqrt{17}+\left(10-2\sqrt{17}\right)\sqrt{17+4\sqrt{17}}-4\sqrt{17+3\sqrt{17}+\left(5-\sqrt{17}\right)\sqrt{17+4\sqrt{17}}+\left(10+2\sqrt{17}\right)\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{2}=\frac{-1+\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+2\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34-2\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+4\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{3}=\frac{-1-\sqrt{17}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}+2\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34+2\sqrt{17}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}-4\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{4}=\frac{-1+\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}-2\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34-2\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+4\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{5}=\frac{-1-\sqrt{17}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}-2\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34+2\sqrt{17}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}+4\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{6}=\frac{-1-\sqrt{17}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}+2\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34+2\sqrt{17}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}+4\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{7}=\frac{-1-\sqrt{17}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}-2\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34+2\sqrt{17}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}-4\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{8}=\frac{-1+\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}-2\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}+2i\sqrt{34-2\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}-4\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{9}=\frac{-1+\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}-2\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34-2\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}-4\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{10}=\frac{-1-\sqrt{17}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}-2\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34+2\sqrt{17}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}-4\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{11}=\frac{-1-\sqrt{17}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}+2\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34+2\sqrt{17}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}+4\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{12}=\frac{-1-\sqrt{17}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}-2\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34+2\sqrt{17}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}+4\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{13}=\frac{-1+\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}-2\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34-2\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+4\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{14}=\frac{-1-\sqrt{17}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}+2\sqrt{17-3\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}+(-5-\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34+2\sqrt{17}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}-4\sqrt{17-3\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+(5+\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{15}=\frac{-1+\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+2\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34-2\sqrt{17}+(-10+2\sqrt{17})\sqrt{17+4\sqrt{17}}+4\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$

$$\omega_{17}^{16}=\frac{-1+\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+2\sqrt{17+3\sqrt{17}+(-5+\sqrt{17})\sqrt{17+4\sqrt{17}}+(-10-2\sqrt{17})\sqrt{17-4\sqrt{17}}}-2i\sqrt{34-2\sqrt{17}+(10-2\sqrt{17})\sqrt{17+4\sqrt{17}}-4\sqrt{17+3\sqrt{17}+(5-\sqrt{17})\sqrt{17+4\sqrt{17}}+(10+2\sqrt{17})\sqrt{17-4\sqrt{17}}}}}{16}$$